

**WEBSITE USER GUIDE WEB-BASED GEOGRAPHIC INFORMATION
SYSTEM FOR FLOOD MAPPING AREA
(FLOOWGIS)**

This document is compiled as a user guide for the web-based Geographic Information System (GIS) website for flood risk analysis developed using Remote Sensing and WebGIS technologies. This guide is intended for general users and disaster management staff to enable them to use the system optimally.

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CHAPTER I

INTRODUCTION

1.1 Background

Floods are one of the natural disasters that have a significant impact on people's lives. Therefore, a system that can provide spatial information quickly, accurately, and easily is needed. This GIS website was developed to visualize and analyze flood risks using remote sensing data.

1.2 Purpose of the Guide

This guide aims to provide technical instructions on how to use the flood risk analysis GIS website so that users can understand the functions, features, and usage flow of the system properly.

1.3 Scope

The scope of this guide includes how to access the website, an explanation of the user interface, and steps for using the main features.

CHAPTER II

OVERVIEW OF THE SYSTEM

2.1 System Description

This web-based Geographic Information System was developed with a client-server architecture. The backend uses Flask as a proxy server connected to Google Earth Engine, while the frontend uses ReactJS and Leaflet for map visualization.

2.2 User Access Rights

This website can be accessed by the general public without authentication. Users can view maps, set layers, and perform spatial analysis.

CHAPTER III

WEBSITE USER GUIDE

3.1 Website Access

Users can access the SIG website through web browsers such as Google Chrome or Mozilla Firefox with a stable internet connection.

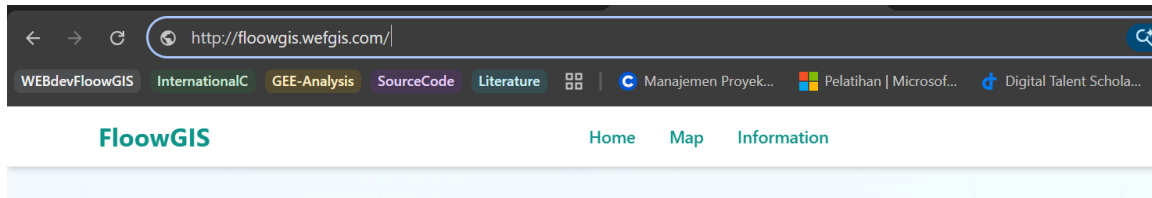


Image of Website Access via Browser

3.2 Landing Page

The landing page is the initial display that presents an interactive map as the main focus of the system. This page is equipped with map navigation, search tools, and layer controls.

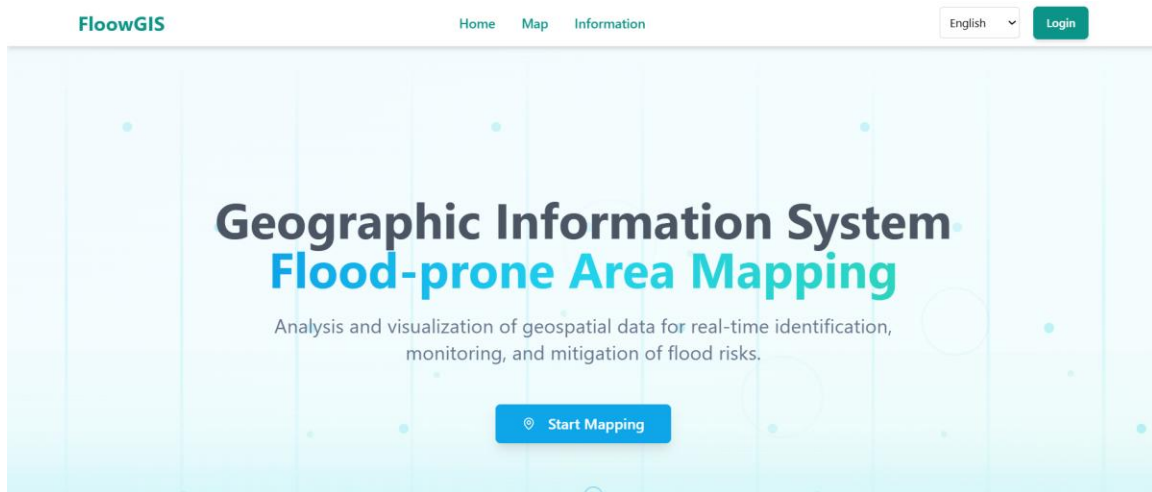


Image Landing Page

3.3 Map Layer Management

Users can enable or disable map layers such as administrative boundaries, public facilities, rainfall, and flood-prone areas through the layer control panel.

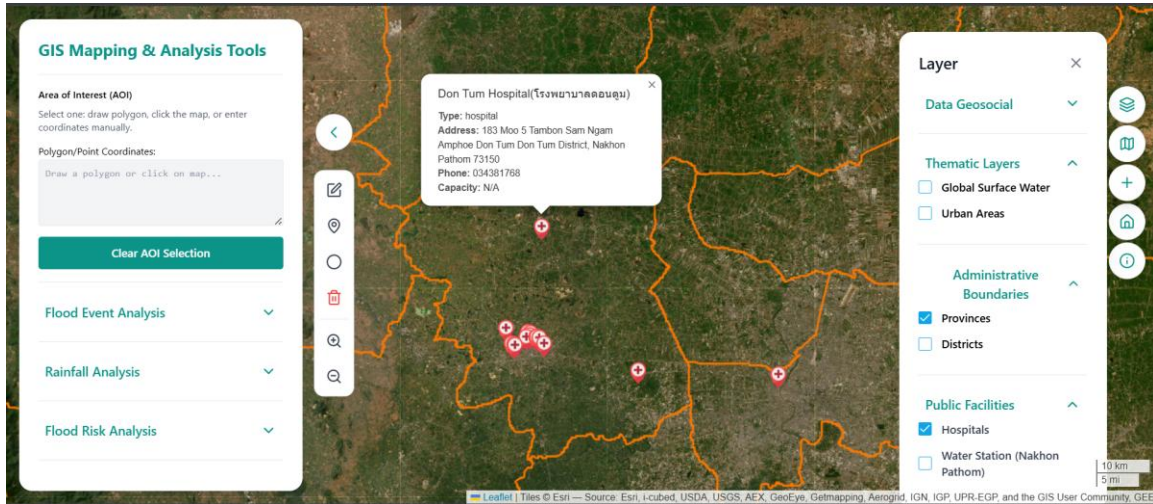


Image of Map Layer Management

3.4 Flood Event Area Analysis

The flood event area analysis feature allows users to analyze historical flood events based on specified areas using polygons or shapefile data.

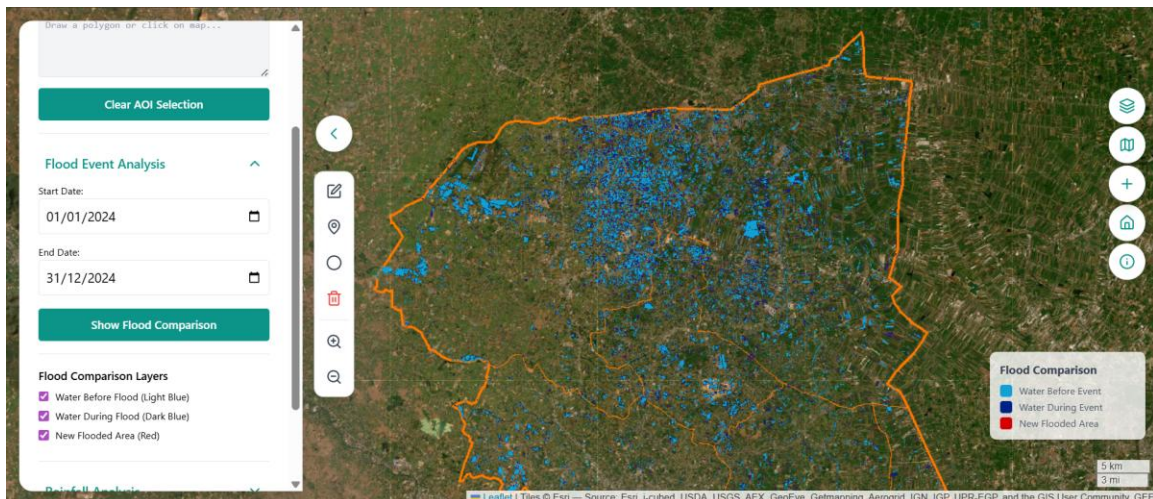


Image of Flood Event Map Analysis

3.5 Rainfall Analysis

The rainfall analysis feature displays historical rainfall data in the form of monthly and daily graphical visualizations to support flood risk analysis. This analysis can be performed in two options, namely monthly and daily data analysis.

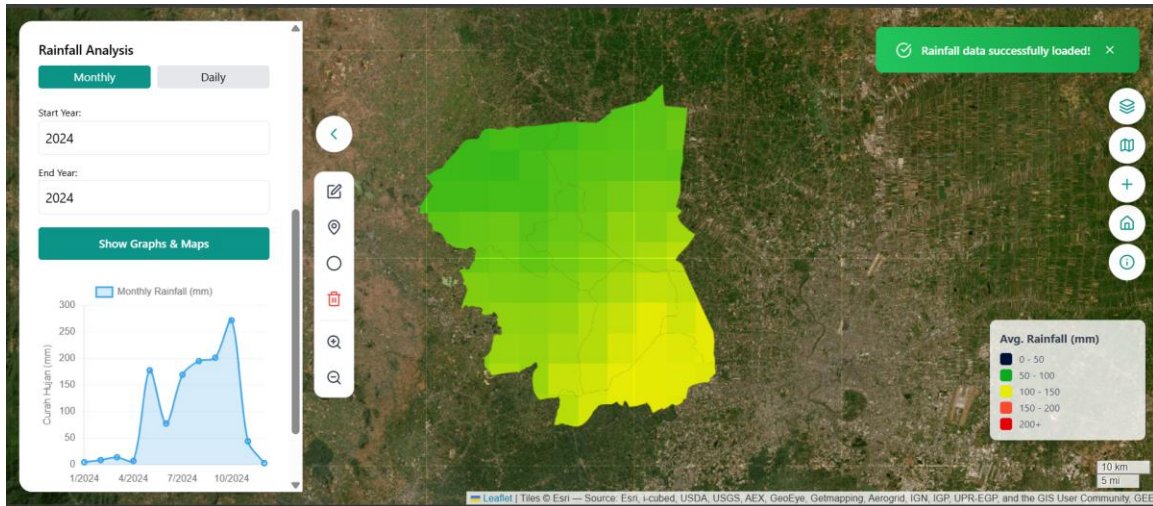


Image of Monthly Rainfall Analysis

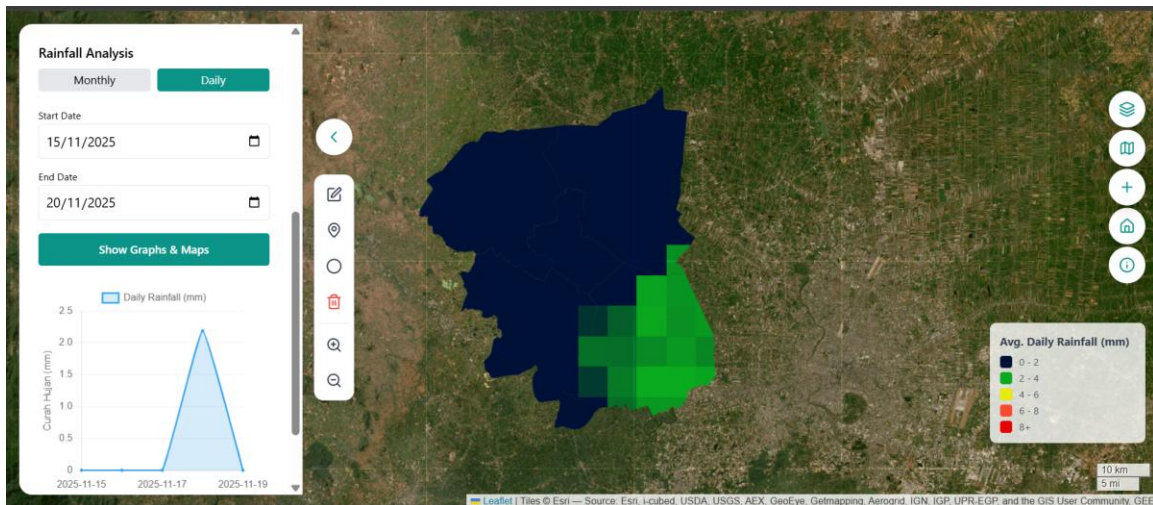


Image of Daily Rainfall Analysis

3.6 Analysis of Flood-Prone Areas

This feature displays a flood risk probability map for the area selected by the user. This information is used as a basis for mitigation decision-making.

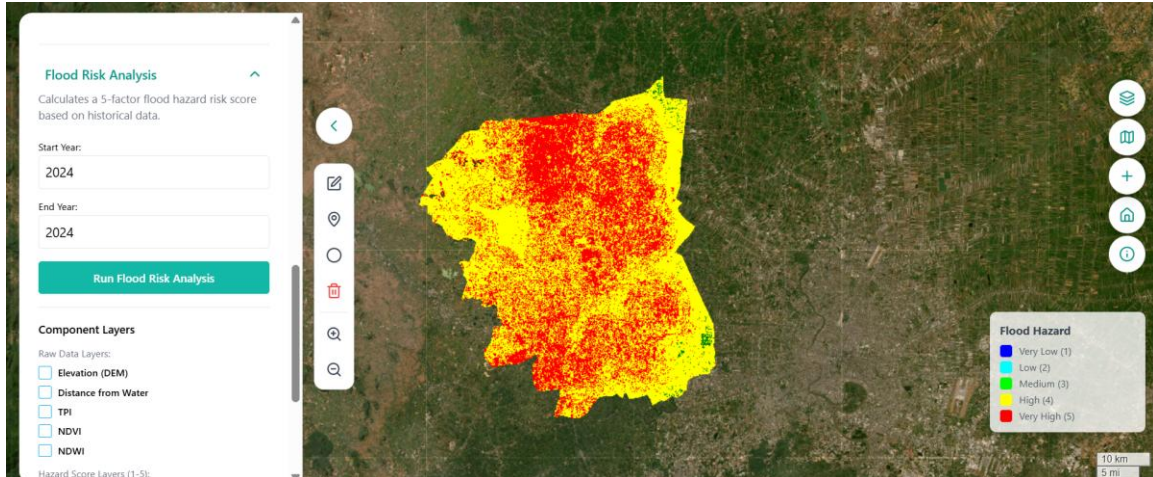


Image of Flood Prone Analysis

3.7 Layer Geosocial Map

This feature presents various layers that support geosocial mapping in the Nakhon Pathom area. This information can be used as a basis for decision-making in mitigation and water management.

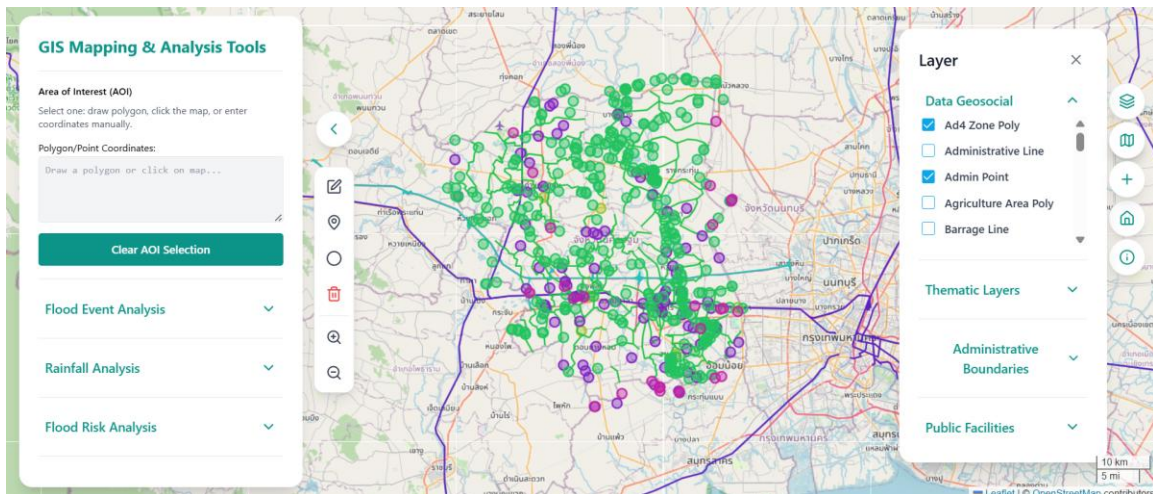


Image of Layer Geosocial Map

CHAPTER IV

CONCLUSION

4.1 Conclusion

This guidebook is expected to assist users in understanding and operating the flood risk analysis GIS website effectively. Further development is recommended to include the integration of mobile platforms and Internet of Things (IoT)-based monitoring systems.